PRINGLE: Security-driven Ontology Framework for Autonomous Driving Systems

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Abstract. Autonomous driving perception models are vulnerable to physical manipulations that create deceptive visual cues and cause unsafe decisions. Vision-Language Models (VLMs) have been proposed as a potential solution; however, they often fail to recognize when perceived information violates fundamental physical, semantic, or contextual constraints. To address this, we introduce PRINGLE, a security-driven ontology framework that enriches VLM-based perception with semantic reasoning capabilities for consistency assessment. Rather than relying just on visual predictions, PRINGLE verifies whether perceived scene elements are consistent with domain knowledge encoded through a threat model-driven ontology. By treating physical manipulations as violations of ontological rules rather than statistical anomalies, PRINGLE detects inconsistent perceptions and provides additional validation to improve safety-critical decisions. We evaluate PRINGLE on traffic light perception under active manipulation in simulated and real-world environments. PRINGLE incorporates ontology-aware training objectives and verifies semantic consistency at inference. Results show that PRINGLE maintains performance while improving resilience to multiple families of sensor injection attacks by detecting inconsistencies between observed and expected traffic light states. These findings highlight the potential of ontology-driven semantic reasoning to strengthen VLM-based perception in safety-critical driving scenarios.

Keywords: Autonomous Vehicle Security · Ontology · VLM

1 Introduction

Physical alterations to roadside infrastructure can generate misleading visual evidence. These alterations therefore represent a serious challenge to perception-based autonomous driving [1–5]. For instance, previous studies [6–8, 5] have shown that visible and infrared light can be directed toward reflective surfaces of traffic lights and traffic signs to produce misinterpretation (e.g., a speed limit confused as a stop sign). Such manipulations may be performed using inexpensive, commercially available devices, including low-power 5mW laser pointers, and can cause advanced traffic light recognition systems, such as Autoware [9],

Apollo [10], and LISA [11], to infer incorrect signal states. These examples reveal a broader limitation of current autonomous driving perception pipelines: learned models may interpret sensor observations as valid even when the resulting visual cues are inconsistent with the semantic and physical conditions of the surrounding environment [12, 13]. Furthermore, current protection strategies in the literature predominantly rely on detecting statistical patterns learned from data or adversarial training, which may fail to capture violations of real-world semantics and physical constraints [5, 14–17].

Recent studies have explored Vision Language Model (VLM) as a means of enhancing autonomous driving perception and decision-making [18, 19]. Nevertheless, their reliability is still under active research, as VLMs have been shown to misinterpret visual evidence and often produce hallucinated outputs, particularly when exposed to manipulated or adversarial inputs [20, 21]. This limitation motivates the integration of structured knowledge and explicit reasoning directly into the perception models so that inconsistent observations can be identified before they influence downstream driving decisions. Specifically, we tackle the following research questions: **(RQ1)** Can ontology-driven semantic reasoning improve the resiliency of VLM-based perception systems? **(RQ2)** Can semantic consistency verification provide a reliable mechanism for assessing the trustworthiness of VLM decisions in safety-critical environments?

To answer these questions we introduce **PRINGLE**, a novel framework which integrate semantic reasoning into VLM-based perception pipelines. The proposed approach, implemented for a traffic light recognition task, improves both model accuracy and resilience to adversarial manipulation by ensuring that visual interpretations remain physically, semantically, and contextually consistent. To support this process, PRINGLE leverages the formal structure of ontologies, providing a reasoning layer that validates sensor observations against predefined domain knowledge and consistency constraints. At runtime, the framework uses this knowledge to apply machine-interpretable physical and regulatory constraints, enabling contextually invalid perceptions to be detected before they influence downstream decisions. Our contributions are summarized as follows:

- Introduction of PRINGLE, a novel ontology-driven reasoning framework that enables the validation of visually plausible but semantically inconsistent or manipulated inputs in VLMs used in autonomous driving.
- The design of a novel, threat model-driven methodology for constructing domain-specific ontologies that treat physical manipulations as ontological and normative violations rather than merely statistical anomalies.
- Evaluation of PRINGLE through a traffic-light perception proof-of-concept case study under active manipulation, demonstrating up to 54.8% decrease in attack success rate in simulation and up to 100% in real-world evaluation against two families of light injection attacks, while maintaining high model performance. These results highlight the potential of ontology-driven seman-

tic reasoning to identify perceptual inconsistencies and improve robustness against manipulated visual inputs [3].

2 Related Work

The integration of explicit semantic knowledge into vision–language perception pipelines for autonomous driving is an emerging research direction. In the following, we discuss the most relevant and closely related works that leverage both Large Language Model (LLM), and VLM-based approaches.

In 2023, Elhafsi et al. [22] introduce the notion of *semantic anomalies*: visually plausible yet contextually implausible scenes, such as an inactive traffic light transported on a truck. They detect these anomalies at runtime by prompting an LLM to assess the contextual plausibility of textual scene descriptions generated from perception outputs. Their results support a core premise of our work: such failures arise from the absence of semantic constraints during interpretation, not from the visual front-end alone. However, their validation relies on the implicit knowledge of a prompted LLM, enforces no formal domain constraints, can only reason over what the upstream perception verbalizes, and targets benign anomalies rather than deliberate manipulation. In contrast, PRINGLE grounds validation in explicit, machine-interpretable ontological axioms derived from physical and regulatory parameters, injects this knowledge already during training, and is designed to augment resiliency against physical adversarial attacks.

In 2024, Tian et al. [18] introduce DriveVLM, which translates visual inputs into structured linguistic scene descriptions that feed LLM-based scene analysis and hierarchical planning, achieving up to double-digit gains over vision-only baselines on long-tail scenarios. Like PRINGLE, DriveVLM relies on a structured linguistic interface between perception and downstream reasoning. However, it uses this interface as-is: it enforces no symbolic constraints on the descriptions, so a manipulated observation propagates unchecked into planning, precisely the gap our inference validator closes. Pan et al. [19] adopt the complementary strategy in Vision Language Planning (VLP): instead of generating language at runtime, they align scene representations and planning queries with semantic embeddings from pretrained language models during training. Although it improves standard planning metrics, the method is limited to adjusting weights and provides no explicit rule to validate perception at inference. In this work, we tackle this challenge by adopting formal domain constraints and an optimized consistency validator.

Recently, Nguyen et al. developed ORCA [23], a training-free failure detector with LLM-generated human-level concepts (i.e., visual attributes) and scores confidence through ordinal ranking of concept activations rather than category-level logits alone. However, these concepts are informal free-text attributes elicited from an LLM rather than axioms grounded in physical and

[3] Trained models, scripts, and experimental materials are provided in the repository: https://github.com/Marc-cn/PRINGLE

regulatory standards, leaving the detector prone to common issues such as misalignment under malicious and ambiguous scenarios [20].

Rule-based validation provides another way to incorporate domain knowledge into deep learning. For instance, C-HMCNN(h) [24] enforces class-hierarchy constraints and arbitrary exclusions on multi-class classification. Ahmed et. al [25] restrict structured predictions to assignments that satisfy predefined symbolic constraints. Finally, ROAD-R [26] defines a fixed set of propositional-logic constraints on a dataset for multi-label classification, preventing conflicting labels from being assigned to the same semantic elements within the scene. However, these methods only prevent contradictions in classification using deterministic rules about valid and invalid relationships between classes; PRINGLE instead incorporates real-world physical constraints as part of a defined ontology, allowing the model to learn semantic consistency as part of the training process.

In this work, we analyze traffic light recognition as a natural stress test for perception consistency: the light is the sole authoritative cue governing right-of-way, a single misread state (e.g., red as green) translates directly into an unsafe maneuver, and VLMs has been used in research to misinterpret such signals under ambiguous and adversarial conditions [20]. The idea of coupling a VLM with a traffic light ontology was first hypothesized by Cardillo et al. [27], who outline a conceptual pipeline. That contribution, however, remains theoretical: no evaluation is reported, making the feasibility of the proposed method unclear. Furthermore, the ontology acts only as a passive downstream filter, while PRINGLE transforms the ontology into an active source of training supervision and provides empirical validation over real-world data.

3 Framework Overview and Threat Model

This work presents PRINGLE, an ontology-driven framework designed to mitigate VLM errors in interpreting safety-critical signals in autonomous driving scenarios. Recent research [20, 7, 5, 8, 6, 4] on AI models has shown that visual learning alone can lead to false interpretation of safety-critical signals, in both benign and adversarial contexts. Although recent VLMs can learn strong associations between images and textual information, they can produce predictions that are visually plausible but semantically inconsistent, contextually impossible, or insufficiently reliable for driving applications.

To tackle this problem, PRINGLE incorporates a security-driven ontology rather than a simple semantic representation of concepts. It applies to a target VLM in two stages as illustrated in Figure 1. During training, the ontology enriches the learned visual representation and contributes additional supervision through semantic attributes and security constraints. During runtime, the same knowledge structure is used in a consistency verification process to validate whether the output complies with domain rules and safety constraints. This modular structure described in Section 4 allows the framework to be adapted to different VLM models, ontologies, and tasks.

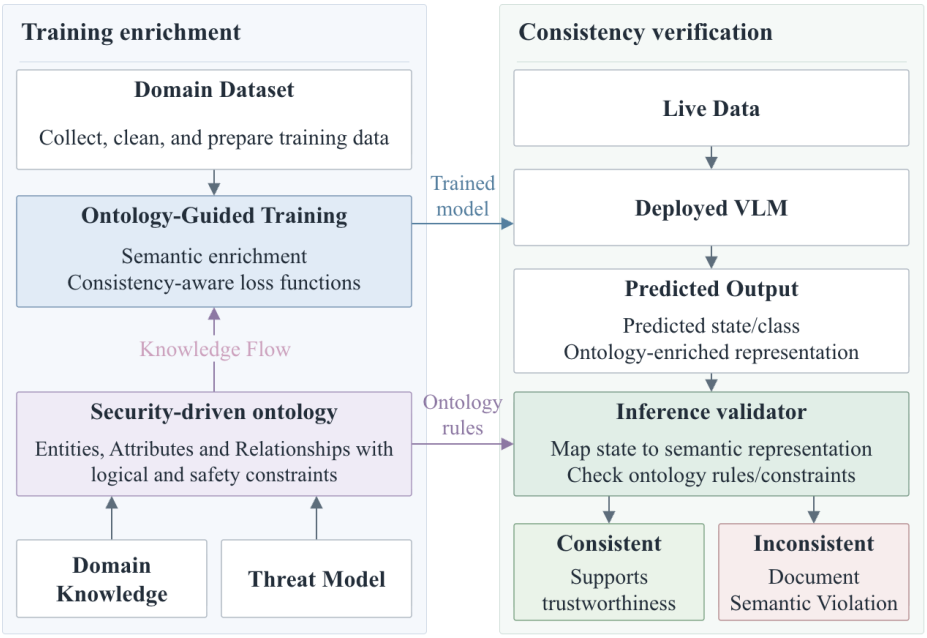


Fig. 1: Workflow integration of PRINGLE into a target VLM model during training and inference phase.

To demonstrate this, we apply PRINGLE to a representative case study of traffic light interpretation in autonomous vehicles under physical adversarial attacks [6, 28], and the framework detect discrepancies between the predicted traffic light state and the actual state under active manipulation, thereby enabling the evaluation of decision trustworthiness and the identification of potential manipulation, as described in Section 5.

3.1 Threat Model

In our threat model, we target traffic light recognition tasks under *physical adversarial manipulation*. In particular, we consider the realistic case scenario where the adversary alters visual cues perceived by onboard camera sensors without tampering with internal software or network traffic or having white-box access to the traffic light model deployed in the vehicle.

Threats. Several attacks in the literature expose vulnerabilities in sensors and AI models governing the detection and interpretation of traffic lights [29, 30, 28, 8]. Here we consider the two main attack families in the literature that induce semantically inconsistent yet visually plausible traffic light observations, which can cause an autonomous vehicle to misinterpret signal states (e.g., mistaking a green status for red) and consequently trigger unsafe automatic decisions, such as activating emergency brakes. These families consist of laser and light injection

attacks that modify the perceived status of the traffic signal by illuminating the traffic light housing containing the red, yellow, and green lenses (called top, middle, and bottom *modules*) to cause misinterpretations, such as illuminating the top module of a traffic light with a red light when the green light is on. These attacks are typically highly successful as the machine learning models cannot distinguish the legitimate light source from the injected one [6, 7, 28].

Attacker capabilities. The adversary is assumed to be remote or in physical proximity to the traffic light, with the ability to use a light source such as laser pointers or portable light sources to perform the attack, as described in previous work. The attacker does not require access to vehicle internals, training data, model parameters, or sensor calibration. No assumptions are made about the attacker's knowledge of the specific vision or AI model in use, making the attack largely model-agnostic. The adversary does not modify traffic light hardware, control logic, or timing. These attacks are localized, transient, and line-of-sight dependent; however, they can be easily pursued with simple off-the-shelf equipment and commercial flashlights or laser pointers.

Attack Methodology. We consider two representative attack methodologies for physical injection attacks developed by Bhupathiraju et al. [6] and Wang et al. [28]. In the first methodology, the adversary injects visible red and green or infrared laser light onto the traffic light lenses. The traffic light module internal reflector scatters the injected beam, causing it to appear to the vehicle's camera as genuinely illuminated by the internal light bulb. In controlled dynamic experiments involving three state-of-the-art traffic-light recognition systems, these attacks achieve 100% success rate in misclassifying green as red and an 86% success rate in misclassifying red as green, at distances of up to 25 m. The second attack methodology [28] exploits the sensitivity of automotive cameras to infrared light that is invisible to the human eye. Rather than relying on the internal reflector of an existing traffic light, the attacker uses infrared LEDs embedded in a physical structure resembling a traffic light. The emitted infrared light is perceived by the camera as a magenta or red signal, causing the vehicle to detect a nonexistent red traffic light. The attack is demonstrated against a Tesla Model 3 at distances of up to approximately 10 m using an infrared light source. Their experiments primarily target false traffic-light detection and also consider broader effects on object perception and navigation (e.g., SLAM [31]).

Testing Scenario. As a target VLM to apply our framework, we chose the CLIP VLM ViT-B/32 multimodal vision–language model (defined in the paper as the "baseline" VLM) developed by OpenAI [32]. This choice stems from preliminary experiments that indicate higher accuracy and generalization compared to ResNet-based alternatives [33] while maintaining substantially lower computational requirements than larger models such as Cosmos-Reason1 [21], which makes this VLM a suitable candidate for on-board vehicle settings. Furthermore, CLIP VLM ViT-B/32 has been used in a variety of classification tasks within the ADAS domain, including traffic sign recognition [34], pedestrian recognition [35], and scene classification [36].

4 Framework Design

PRINGLE couples a VLM-based perception stack with an ontology-based reasoning layer to validate whether object interpretations are not only visually plausible, but also physically and semantically consistent before they can influence downstream decisions. The pipeline is organized into two phases shown in Figure 1: a *Training Enrichment* phase that uses structured domain knowledge derived from a security-driven ontology, and a *Consistency Verification* phase that performs runtime validation through formal reasoning. This section describes the overall framework design and case study application on traffic light interpretation.

4.1 Ontology Design

In PRINGLE ontologies are used to provide a machine-interpretable specification of domain constraints to guide the target model.

Our approach adopts an explicit security-driven conceptualization of the reference domain, in which existing semantic resources (e.g., taxonomies, standards, and domain vocabularies) and threat models are analyzed jointly. Focusing on the concepts and relationships required for a target threat model instead of modeling all possible domain entities and interactions enables the ontology to directly support the identification and analysis of security-critical conditions. In our case study, we use a semi-automatic knowledge extraction process that combines Knowledge Graphs and BERT fine-tuned for Natural Language Inference (BERT-NLI) [37] to identify the main candidate concepts of the domain and semantic relationships between them. After verification by our domain expert, we construct the associated ontological meta-model using structured standard ontology languages like OWL [38]. This meta-model is then used to create new domain-specific loss functions to incorporate into model training and consistency verification.

In our case, we formalize the structural and operational semantics of a physical traffic light system from a security perspective based on our threat model. In our case, signal intensity, color, and module status represent the fundamental elements exploited by an attacker. In light of this, from the knowledge sources two key standards governing the colorimetric and photometric characteristics of traffic lights are considered: CIE 1931 [39] and EN 12368:2015 [40].

For this proof-of-concept design we consider a standard vertical traffic light system with three modules as most common target of manipulation [8, 28, 7, 6]. This ontology can be easily extended to include other traffic light configurations, states, and parameters, or include different objects like billboards and signs, by adding new concepts and semantic relationships to the constructed meta-model.

We define the class `SignalModule` with three subclasses, each corresponding to one module position (Top, Middle, Bottom). Each class is related to the object property `controls` (inverse property of `isControlledBy`) to the class `SignalState` that represent the operative state of a module: green, yellow, red,

redyellow (meaning red and yellow state on simultaneously), and off-state indicating when the traffic light is disabled. With respect to other photometric parameters, based on the aforementioned standards, we also defined the range of *luminous intensity* expected for an active traffic light module, i.e., the light intensity emitted by the source in a given direction, expressed as an observer-independent property measured in candela (cd) at a reference axis:

$$\texttt{SignalModule} \sqsubseteq \exists\, \texttt{hasLuminousIntensity}.[200, 800]$$

The main axioms defined in the ontology are the following:

- **Positional compatibility**: e.g.,
  - $\texttt{RedState} \sqsubseteq \forall\, \texttt{isControlledBy}.\texttt{TopModule}$
- **Cardinality constraints**: e.g.,
  - $\texttt{TrafficLight} \sqsubseteq = 3\, \texttt{hasSignalModule}.\texttt{SignalModule}$
- **Datatype restrictions** following the CIE and EN 12368:2015 permitted ranges: e.g.,
  - $\texttt{RedState} \sqsubseteq \exists\, \texttt{hasCIE_x}.[0.67,\ 0.70]^{\texttt{xsd:decimal}}$;
  - $\texttt{RedState} \sqsubseteq \exists\, \texttt{hasDominantWavelength}.[620.0,\ 630.0]^{\texttt{xsd:decimal}}$
- **Shape constraint**: e.g.
  - $\texttt{SignalModule} \sqsubseteq \forall\, \texttt{hasShape}.\texttt{CircularShape}$

Moreover, we populate the ontology with valid and invalid individuals i.e., the specific instances of the classes of the ontology (e.g., "GreenState_A01" is one instance the "Green State" class, "BottomModule_A01" is one instance of the "Bottom Module" class), and we verify the absence of inconsistencies with two DL-reasoners, HermiT [41] and Pellet [42].

The resulting ontology, comprises 15 classes, 9 object properties and 6 data properties, for a total of 191 axioms (91 logical), 4 annotation properties and 13 individuals. An extract of the ontology graph is shown in Figure 2.

4.2 Training Enrichment

During this phase, the framework jointly optimizes the target model using the primary task objective and ontology-derived semantic constraints, enabling learning through multi-task supervision (detailed in Section 5). This approach allows the target model to (i) separate visually similar but semantically distinct states, (ii) encode physically grounded cues, such as colorimetric ranges and intensity in our case study, and (iii) remain compatible with the semantic constraints. Rather than learning purely from correlations, the ontology acts as a source of *structured regularization*.

Semantic enrichment. During training, the ontology is parsed as a Resource Description Framework (RDF) graph [43] to identify properties that are explicitly instantiated and sufficiently populated. In our case study, for each canonical traffic light state, the corresponding ontology node is queried to retrieve its associated attributes (e.g., control module, chromaticity ranges, wavelength intervals). Within this phase, the ontology is integrated into the model through

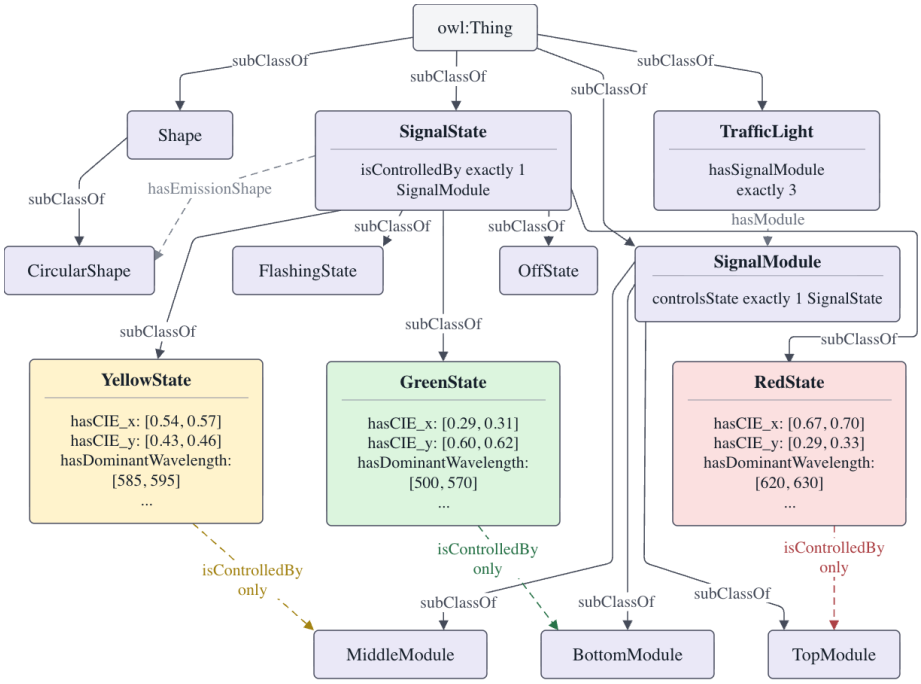


Fig. 2: Graph-based overview of the ontology used for traffic light interpretation.

ontology-aware loss functions, which guide training by enforcing consistency with the represented semantic and physical constraints. During inference, the model produces both its primary prediction and an ontology-derived message with relevant semantic attributes associated with the output, which is used during the consistency verification phase. This output message is designed to support decision interpretability and inform about inconsistencies related to the ontology.

Multi-task supervision through additional loss functions. In machine learning, a loss function quantifies the discrepancy between model predictions and ground-truth supervision signals, and provides the optimization objective used to update model parameters during training [44]. In PRINGLE, we modify the VLA model loss to jointly enforce adherence to the photometric qualities of traffic light modules and consistency between these photometric properties and the predicted state, penalizing any prediction that violates either constraints.

4.3 Consistency Verification

During runtime, the real-time camera inputs are processed by the trained model with PRINGLE to produce a predicted object state together with the enriched description integrated by our framework. This information is then passed to a *inference validator* which verifies that the predicted state and extracted descriptors do not contradict the ontology constraints. If the verification succeeds, the

output is treated as coherent and admissible; otherwise, the validator raises an anomaly. Specific implementation details are described in Section 5.

From a cybersecurity perspective, the inference validator can be interpreted as a perception-level uncertainty evaluator: instead of attempting to detect attacks through visual heuristics, it validates whether the perception output respects formalized domain rules and physical plausibility constraints. The consistency verification therefore provides information that can be consumed by downstream planning and control modules, enabling fail-safe behaviors when perception becomes suspicious.

5 PRINGLE Implementation on Traffic Light Recognition

As describe in Section 3.1, we target the CLIP ViT-B/32 VLM (called CLIP in the rest of the paper for simplicity), an OpenAI model pretrained on approximately 400 million image–text pairs. This baseline model is initialized with the given pretrained CLIP weights and fine-tuned for traffic light classification by replacing the generic zero-shot prompt matching with a task-specific classification head, enabling the model to learn from discrete signal states with supervised labels without any text or prompt input.

In this phase, we leverage the DriveU Traffic Light Dataset (DTLD) [45], a large-scale, publicly available benchmark for traffic light research comprising roughly 40,000 annotated images with detailed pictogram labels that is widely used in recent traffic-light detection and recognition studies [46, 47]. Within these, there are approximately 292,000 traffic light bounding boxes. We remove any boxes with an unknown state, as well as any occluded or non-front-facing traffic lights, to provide clean, labeled traffic light images for training. Additionally, we exclude all traffic lights that do not follow a vertical three-module configuration, as the evaluated attacks have been validated exclusively on this traffic light type, obtaining 97,918 valid traffic light images.

Training. A linear classification head maps the CLIP image embeddings to the five-class canonical traffic light state space: *red*, *yellow*, *green*, *red_yellow*, and *off*. Training with the DTLD dataset follows a stratified 80/20 train–validation split and is executed with a fixed random seed to ensure reproducibility. Images are then resized to 224×224 RGB resolution and normalized using CLIP-specific statistics. To mitigate the pronounced class imbalance of the dataset without altering its composition, we apply a weighted sampling strategy during mini-batch construction, while optimization relies on AdamW [48] with cosine learning-rate scheduling, selected after a preliminary hyperparameter evaluation. In validation on the DTLD dataset, CLIP achieves a true positive rate (TPR) of 90.8% and a true negative rate (TNR) of 99.2% after 50 epochs, with a TPR and TNR score of over 91.2% on the green, red, yellow, and red-yellow states. The TPR of the off state is lower, at 71%. This reduced performance on the off state is consistent with other models trained on the DTLD dataset. For instance, Mentasi et al. [49] report off state average precision below half that of the other four classes across four different models trained for joint detection and classification. The

overall Macro-F1 of the CLIP model is 0.925, in line with typical traffic light recognition models [7].

5.1 PRINGLE Integration

The perception task is decomposed into multiple objectives derived from the ontology RDF graph, following the procedure detailed in Section 4.2. In our case study, learning is driven by the following weighted multi-task loss:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{attr}} + \lambda_2 \mathcal{L}_{\text{con}} + \lambda_3 \mathcal{L}_{\text{rank}} + \lambda_4 \mathcal{L}_{\text{card}} \left(\mathcal{L}_{\text{color}} + \mathcal{L}_{\text{intensity}}\right)$$

where each term supervises a distinct prediction head and captures a different modeling objective. The λ_x coefficients regulate the relative influence of ontology-induced supervision with respect to the primary classification objective, and are chosen via optimization alongside a hyperparameter search. The loss functions are detailed as follows:

- $\mathcal{L}_{\text{cls}}$ is the standard cross-entropy classification loss on traffic light signal states (e.g., red, yellow, green). This is the *base perception task* of the baseline VLM, and represents the same objective used in conventional traffic light recognition pipelines [7].
- $\mathcal{L}_{\text{attr}}$ supervises the prediction of discrete ontology-derived symbolic attributes of dominant wavelength, CIE X value, and active module. Although also formulated as a classification loss, it differs from $\mathcal{L}_{\text{cls}}$ as it targets the *semantic properties* rather than the primary signal state. This separation allows the model to learn disentangled representations aligned with domain knowledge.
- $\mathcal{L}_{\text{con}}$ is a contrastive alignment loss that enforces consistency between visual embeddings and ontology-aware textual embeddings. This term encourages the shared embedding space to reflect structured semantic relationships defined by the ontology. This embeds the same derived symbolic attributes as the attribute loss in a structured sentence, for example:

  ```
  A photo of a traffic light showing the green state,
  where the signal attributes are: isControlledBy is
        BottomModule, hasCIE_x is 0.15 to 0.20,
       hasDominantWavelength is 505nm to 520nm.
  ```

- $\mathcal{L}_{\text{rank}}$ is a ranking loss on the relative brightness of each module, ensuring coherence between individual traffic light modules and the predicted state. For each predicted state, we enforce that the active traffic light module(s) have a higher relative brightness than the inactive modules. For example, if the predicted state is a red light, then the top module should be brighter than both the middle and bottom modules.
- $\mathcal{L}_{\text{card}}$ is a joint loss term on the photometric parameters derived from the ontology. $\mathcal{L}_{\text{color}}$ is a hinge loss on the 1931 CIE color space for each module state, starting from the expected color of each state (e.g., $\texttt{GreenState} \sqsubseteq \exists\, \texttt{hasDominantWavelength}.\{505.0\}^{\texttt{xsd:decimal}}$). These color ranges expand to

capture 95 % of the predicted values of a given state, allowing the model to naturally distinguish color using the ontology as reference. $\mathcal{L}_{\text{intensity}}$ first normalizes the relative brightness of each module with softmax. These are then mapped to luminous intensity values, which are supervised by SmoothL1 to push predictions towards the midpoint of each luminous intensity range defined by the ontology. $\mathcal{L}_{\text{intensity}}$ and $\mathcal{L}_{\text{color}}$ provide the active modules and the color of each active module, respectively. $\mathcal{L}_{\text{card}}$ uses the predictions from these two functions to perform a cross entropy loss on the five valid states, ensuring that module features adhere to the cardinality of the predicted state.

The model's overall performance before and after PRINGLE integration after 50 training epochs preserves the model's classification accuracy, achieving a Macro-F1 score of 0.924. This demonstrates that the loss functions introduced by PRINGLE do not degrade the model performance.

Consistency Verification. During runtime, the model enriches output with the prediction head of $\mathcal{L}_{\text{color}}$ and $\mathcal{L}_{\text{intensity}}$, proving predicted luminous intensity and CIE color values for each individual module within the traffic light. This enhanced output is provided to the inference validator, which then maps these predicted values to the constraints defined by the ontology model. If these properties do not align with the rules and constraints of the predicted state, the inference validator flags it as uncertain and provides a text output explaining why, as illustrated below:

```
UNCERTAIN: Cardinality of predicted state (red_yellow) is violated.
Active modules: top, middle, bottom. Colors: top=red, middle=yellow,
bottom=other. Reasons: on-mask {top, middle, bottom} is not a legal
traffic-light pattern; bottom is other (expected green). Intensities
(on if I≥200): top=350.4[on], middle=324.7[on], bottom=350.0[on] CIE
xy: top=(0.698,0.290)→red, middle=(0.625,0.402)→yellow,
bottom=(0.560,0.311)→ other
```

6 Evaluation

Using PRINGLE, we perform a comprehensive evaluation under adversarial attacks to determine the capability of our enriched VLM to withstand active manipulation. We compare the performance of both the baseline and the enriched model under the two injection attack families described in Section 3, using visible lasers and invisible (IR) light. This evaluation is performed using the synthesis methodology defined by Bhupathiraju et al. [6], where we take real-world images of traffic light modules under an injection attack collected in our laboratory, mask the traffic light module where the attack takes place, and overlay them onto the DTLD-provided traffic light crops, adjusting for local luminance. Finally, we perform validation of our framework by performing each attack in the real world, during daytime and nighttime conditions.



Fig. 3: Our real-world collected samples of IR light injection over legitimate red and green traffic light status collected in both day and nighttime conditions.

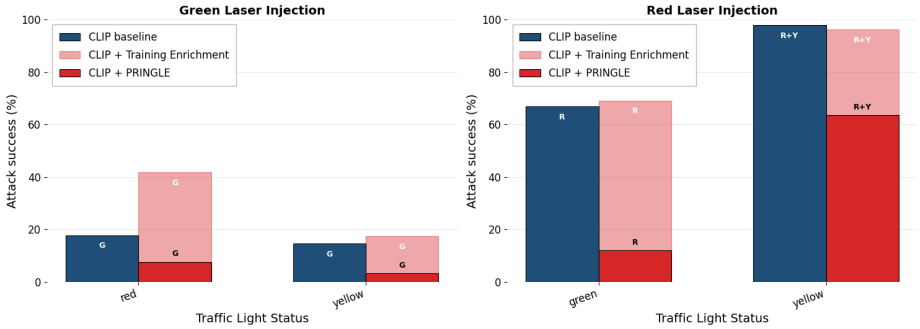


Fig. 4: Attack success rate of green and red laser injection on both baseline (CLIP) and with PRINGLE (CLIP + PRINGLE), including the predicted state (G - green, R - Red, R+Y - Red-Yellow, Y - Yellow). Light red represents the training enrichment without consistency verification. The x-axes represent the legitimate traffic light status during the attack.

6.1 VLM Under Attack

To evaluate CLIP's performance under laser injection and IR light source attacks, we adopt the same experimental setup used in prior studies [7, 28]. Specifically, we train the baseline model on the benign dataset for 50 epochs and select the best-performing model (F1 of 0.925). Then, for each attack instance, we consider the attacker's best-case scenario by selecting the configuration that achieves the highest attack success rate. An attack is deemed successful when the injected signal causes the model to alter its predicted traffic light state, for example, by misclassifying a green signal as red or vice versa. Therefore, attack success rate is calculated as the percentage of images previously classified correctly by the model that now misclassify due to the attack. Green and red laser injection attacks can only be applied to their corresponding traffic-light modules, as a green (red) laser cannot realistically affect a red (green) illuminated module. In contrast, IR light injection can be directed at any of the three non-active modules because infrared illumination is invisible to human observers but can

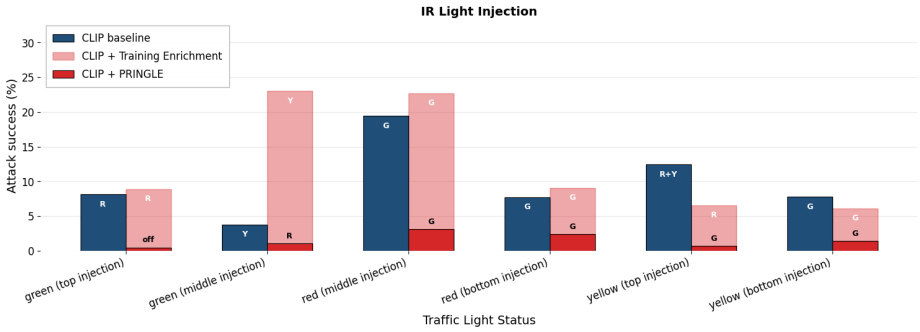


Fig. 5: Attack success rate of IR light injection on both baseline (CLIP) and with PRINGLE (CLIP+PRINGLE), including the predicted state (G - green, R - Red, R+Y - Red-Yellow, Y - Yellow). The x-axes represent the active traffic light module during the injection and the location of the injection (top, middle, and bottom module). Light red represents the training enrichment without consistency verification.

still be captured by the camera sensor and influence the model's prediction, as shown in Figure 3.

Results and Observations. As shown in Figure 4 (blue bars), the baseline CLIP VLM appears inherently more robust to green laser injection than the deep learning models evaluated in prior work [7]. Specifically, the attack success rate is only 18% for green laser injection, whereas the model remains highly susceptible to red laser injection, which achieves an attack success rate of 67%.

Similarly, the IR light injection attack, as shown in Figure 5 (blue bars), has a maximum attack success rate of 19% achieved on injection into the middle module on red benign, and a minimum attack success rate of 4% on injection into the middle module on green benign. When the malicious IR light is directed at the middle or bottom module, the model predominantly misclassifies the signal as green rather than as red-yellow or yellow, respectively. This behavior suggests that the model relies more heavily on the spatial location of the illuminated module rather than color. A similar bias was previously reported by Bhupathiraju et al. [7] for conventional deep learning models. Our results demonstrate that this phenomenon also extends to VLMs, indicating that they exhibit a comparable preference for positional information when subjected to IR injection attacks.

6.2 PRINGLE Evaluation

As shown in Figures 4 and 5 (red bars), PRINGLE substantially improves resilience, resulting in a lower attack success rate across all injection attacks. Specifically, the attack success rate decreases by a minimum of 10% for green laser injection on red traffic light state and up to 54.8% for red laser injection on green. For IR injection attacks, the overall attack success rate decreases by

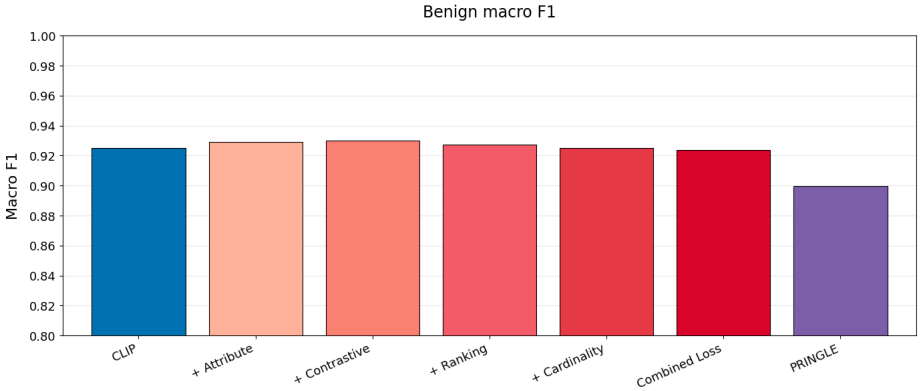


Fig. 6: Macro-F1 of the baseline CLIP model and the loss function components of PRINGLE. Each individual part of the loss function, indicated by the +, is added independently to the baseline CLIP model and evaluated. The combined loss includes all parts of the loss function, and PRINGLE includes the inference validator.

at least 8.36% across all injection attack types, reaching nearly zero (0.5%) in injection into the top module on a green traffic light.

Corner cases. We also analyze two corner cases that, while not considered as inconsistencies from the ontology perspective, can nevertheless be induced by injection attacks. The first case is the red-yellow state in the DTLD dataset. In several European countries, including Germany and the United Kingdom, the simultaneous display of red and yellow signals represents a transitional state indicating that the traffic light will soon turn green and that drivers should prepare to proceed. In contrast, this state is generally absent in the United States and other countries, where traffic signals typically transition directly from red to green. As a result, the semantic meaning and relevance of the red-yellow state depend on the geographical region and traffic signaling conventions under consideration. In our case, PRINGLE treats the red-yellow configuration as a valid state according to the ontology. Consequently, a red laser injection applied during a benign yellow state does not trigger the inference validator. Nevertheless, the overall attack success rate still decreases by 34%, indicating that the model has become more resilient to this attack scenario. The second case is that the traffic light is off and only malicious injection is applied. Similar to the red-yellow scenario, this condition does not violate the ontology, and therefore it passes through consistency verification. PRINGLE still reduces the overall attack success rate by a minimum of 0.4%, to a maximum of 56%, demonstrating improved robustness even in cases where the verification is not applicable.

Ablation Study. To understand the impact of the training enrichment alone, we evaluate the model performance after adding each component of the loss function independently. As shown in Figure 6, attribute, contrastive, and ranking

loss each maintain or slightly improve the Macro-F1 of the baseline model, as they help align the model's embedding space to be semantically consistent with the ontology. Combined, our complete loss function maintains the performance of the baseline model (F1 of 0.924) and increases the overall TPR from 90.8% to 92.1% while maintaining TNR at 99.1%. Applying the inference validator only slightly degrades the overall performance (F1 of 0.90).

We also study the overall attack success rate without applying the inference validator (light red bars in Figures 4 and 5). The new loss functions generally decrease (5.8% max decrease) or show similar attack success rates as the baseline model, with a few exceptions in certain cases. For example, green laser injection increases the attack success rate from 17.8% to 41.8% over red traffic light status. We hypothesize this is due to the higher apparent brightness of the green laser beam compared to red and IR lasers in our synthesized images, which is captured by the ranking loss function.

IR light injection into the middle module instead increases the attack success rate from 3.8% to 23% in the case of green status and from 19.4% to 22.7% in case of red status. Even in this case, this suggests intrinsic bias in favor to spatial location rather than color. The application of the validator allows to flag these semantic inconsistencies, substantially decreasing the attack success rate.

These results demonstrate that PRINGLE training enrichment and the inference validator both concur in increasing model resiliency.

6.3 Real World Validation

We evaluate the real-world performance of PRINGLE by reproducing the injection attacks in outdoor traffic-light scenarios and measuring the resulting reduction in attack success rates. To emulate the attack methodology proposed by Wang et al. [28], we use a UniqueFire UF-2002D infrared flashlight and capture images using a Leopard automotive camera LI-USB30-AR023ZWDR [50]. The data are collected at distances of 24 ft (7.32 m), 36 ft (10.97 m), and 48 ft (14.63 m) from the traffic light under both daytime and nighttime conditions. For each distance and attack configuration, ten images are acquired, resulting in a total of 600 images across the ten IR injection attack configurations. The collected images are subsequently cropped following the procedure described in Section 5. Examples of the IR injection attack are shown in Figure 3.

Across the green, red, and yellow traffic-light states, PRINGLE reduces the overall attack success rate from 51.11% to 14.44%, corresponding to an overall reduction of 36.66%. The reduction is also substantial for the red-yellow and off states, where the attack success rate decreases from 85.41% to 15.41%, representing a 70% reduction. In these cases, PRINGLE is able to identify violations in traffic-light chromaticity even when the spatial position of the injected signal corresponds to an otherwise valid state.

For the red and green laser injection attacks, PRINGLE reduces the attack success rate to 0% for green laser injection on a benign red traffic light state and for red laser injection on a benign yellow state. For both attack types, we also observe an average reduction of 48.34% in attack success rate when the traffic

light is off. For all remaining attack configurations, we were unable to achieve a successful attack against the baseline CLIP model. We hypothesize that this result is primarily due to variations in environmental illumination in outdoor settings, which reduce the effectiveness of our attack setup. In particular, the limited optical power of the 5 mW laser pointer used in our experiments may have been insufficient to reliably alter the perceived traffic-light state under varying lighting conditions, as also shown in the literature [7]. Overall, the results demonstrate that PRINGLE maintains strong resilience in real-world attack scenarios, achieving an average of 50% reduction in attack success rate compared to the baseline CLIP model.

7 Discussion

The outcomes of the analysis described in this work demonstrate that semantic frameworks based on ontologies can act as a *protection layer* for perception systems in safety-critical scenarios. Our evaluation shows that while the data-driven strategy typical for VLMs continue to be prone to overreacting to visual ambiguity, our enrichment framework filters implausible interpretations and reintroduces domain knowledge. This proof-of-concept demonstration on traffic light interpretation confirm PRINGLE capability to protect against malicious manipulations while maintaining a high degree of selectivity for genuine traffic signals. This indicates that combining AI perception with structured knowledge is a promising direction toward resilient and trustworthy safety-critical recognition systems. Besides, the proposed pipeline PRINGLE is designed for real-world deployment, providing interpretable decision-making and a clear path for extension to additional traffic-infrastructure perception tasks beyond traffic light recognition.

Generalization. PRINGLE is designed to be conceptually general and not inherently tied to a single perception task or sensing modality. It separates data-driven perception from ontology-based validation through a well-defined interface, enabling the reuse of the same pipeline structure with different VLM models and ontologies. While traffic light manipulation is used as a representative and safety-critical case study, the methodology can be extended to other authoritative roadside objects (e.g., road signs, pedestrian signals) and, more broadly, to perception tasks where physical or semantic plausibility can be formally specified. Generalization is primarily achieved by replacing or extending the domain ontology and adapting the set of validated attributes and axioms, while preserving the training and operational workflow. As a result, extending PRINGLE does not require redesigning the perception pipeline, but rather enriching the knowledge base and the associated constraints. In principle, the ontology could also support automatic attack detection that generates a semantic anomaly.

Limitations and Future Work. A key limitation of PRINGLE lies in the constrained scope of the evaluated case study. The ontology-driven validation is currently instantiated for a single, defined scenario of traffic light interpretation, and primarily captures static structural and photometric constraints derived

from available standards. In addition, the VLM is trained at best effort using a fixed backbone, dataset configuration, and empirically selected hyperparameters. While the achieved accuracy is high and sufficient to demonstrate the effectiveness of our framework, the model is not claimed to be optimal. Finally, the construction of the loss function treats color, module brightness, and scene brightness as independent features within the scene. This choice in construction allows each part of the loss to be separately constructed from the ontology, however it might overlook subtle relationships of physical phenomena. Despite these limitations, our proof-of-concept study highlights the potential for semantic reasoning adoption on VLM-based technology to be employed in safety-critical driving settings, even in the presence of attackers.

Future work can explore more sophisticated tasks, including the incorporation of temporal aspects into both perception and reasoning processes. The ontology can be extended to capture temporal consistency by modeling valid state transitions over time and verifying that observed sequences follow expected signaling patterns.

7.1 Ethical considerations

Real-World Evaluation. The attack evaluation in our outdoor scenario was conducted in controlled environments adhering to the safety policy of our institutions. All the laser experiments in this work were conducted using appropriate protection, using 5 mW red and green laser pointers for slide presentations (Class 3R and Class 2). The infrared light experiments were conducted using a commercial flashlight.

Use of AI-based Large Language Models. In this work, we used OpenAI and MS Copilot models strictly for improving writing clarity and correcting grammatical errors.

8 Conclusions

The work presents a practical framework integrating an ontology-based reasoning layer to identify semantically implausible scenarios in autonomous driving, including traffic-light signal manipulation. This hybrid approach maintains perception accuracy and improves overall resiliency compared with a purely visual heuristic model. Our evaluation achieves an overall decrease in attack success rate of up to 54.8% with the application of PRINGLE. The outcomes of this study demonstrate that semantic reasoning can improve VLM-based perception systems in safety-critical scenarios. The proposed framework is designed to be adaptable to different VLMs and ontologies by design, providing interpretable decision-making and a clear path for extension to additional traffic-infrastructure perception tasks.

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